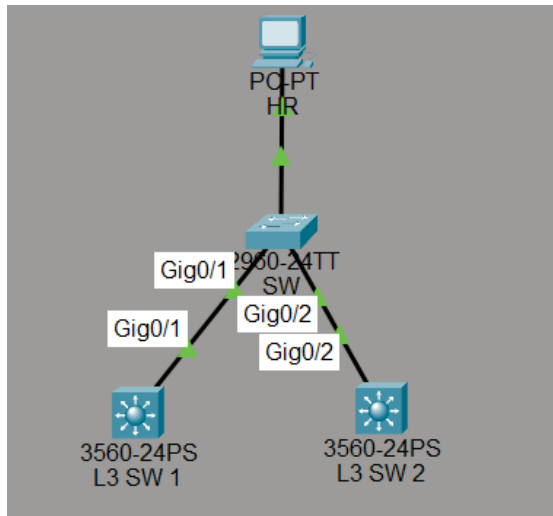


MINI PROJECT 8

HSRP (First Hop Redundancy) + Layer-3 Switching + VLAN Gateways

DIAGRAM:



DESCRIPTION:

This project explains how HSRP (First Hop Redundancy Protocol) provides a virtual default gateway for hosts in a VLAN to ensure high availability.

STEP 1: Create VLANs on SW 1,SW2:

Verify existing VLANs from Previous projects

STEP 2: Assign Access Ports to VLANs on SW1 & SW2:

SW:

```
interface fastEthernet0/1
switchport mode access
switchport access vlan 10
exit
```

STEP 3: Configure SVI IPs on Both L3 Switches

```
L3 S1,
interface vlan 10
ip address 192.168.10.2 255.255.255.0
no shutdown
exit
```

```
L3 S2,
interface vlan 10
ip address 192.168.10.3 255.255.255.0
no shutdown
exit
ip routing
```

STEP 4 : Configure HSRP (Virtual Gateway):

```
L3 switch1(active high priority):
interface vlan 10
standby 1 ip 192.168.10.1
standby 1 priority 110
standby 1 preempt
exit
sw2 & L3switch(standby):
interface vlan 10
standby 1 ip 192.168.10.1
standby 1 priority 100
exit
```

STEP 5: Configure IP Addressing on PCs:

PC	VLAN	IP Address	Virtual Gateway
PC-HR	10	192.168.10.10/24	192.168.10.1

STEP 6 Verification & Failover Test:

```
Switch#show standby brief
                P indicates configured to preempt.
                |
Interface    Grp  Pri  P State      Active        Standby        Virtual IP
Vl10         1    110  P Init       unknown       unknown        192.168.10.1
Switch#
```
